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PLC

Introduction

1. PLC Fundamentals



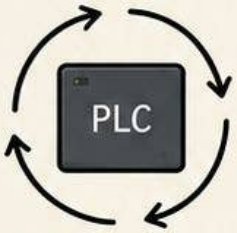
2. PLC Architecture



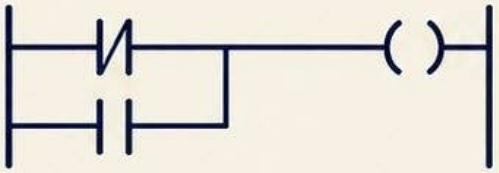
3. PLC Inputs & Outputs



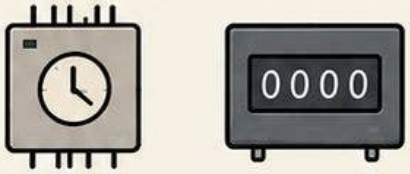
4. PLC Scan Cycle



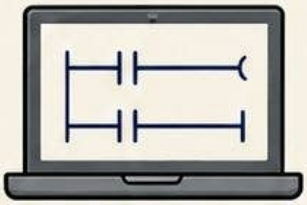
5. Ladder Logic Fundamentals



6. Timers & Counters



7. Programming Languages



LD
FBD
ST
IL

8. Communication Protocols



9. HMI & SCADA



10. Applications & Troubleshooting



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Topic 1

PLC Fundamentals

1. Definition

PLC (Programmable Logic Controller) is a digital electronic device used to automate industrial processes and machines. It monitors inputs, executes a program and controls outputs.



2. Need for PLC

- To replace hardwired relay logic.
- To increase reliability and reduce maintenance.
- To provide flexibility in control.
- To handle complex automation with ease.

3. Features

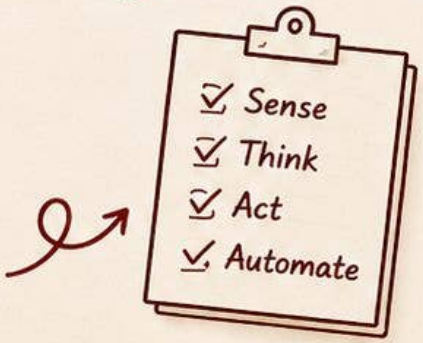
- Programmable and reconfigurable
- High reliability and noise immunity
- Compact size
- Fast operation
- Easy troubleshooting & diagnostics
- Supports communication networks

4. Advantages

- Reduces wiring and installation time
- Easy to modify and expand
- Improves system reliability
- Reduces downtime
- Cost effective in long run

5. Applications

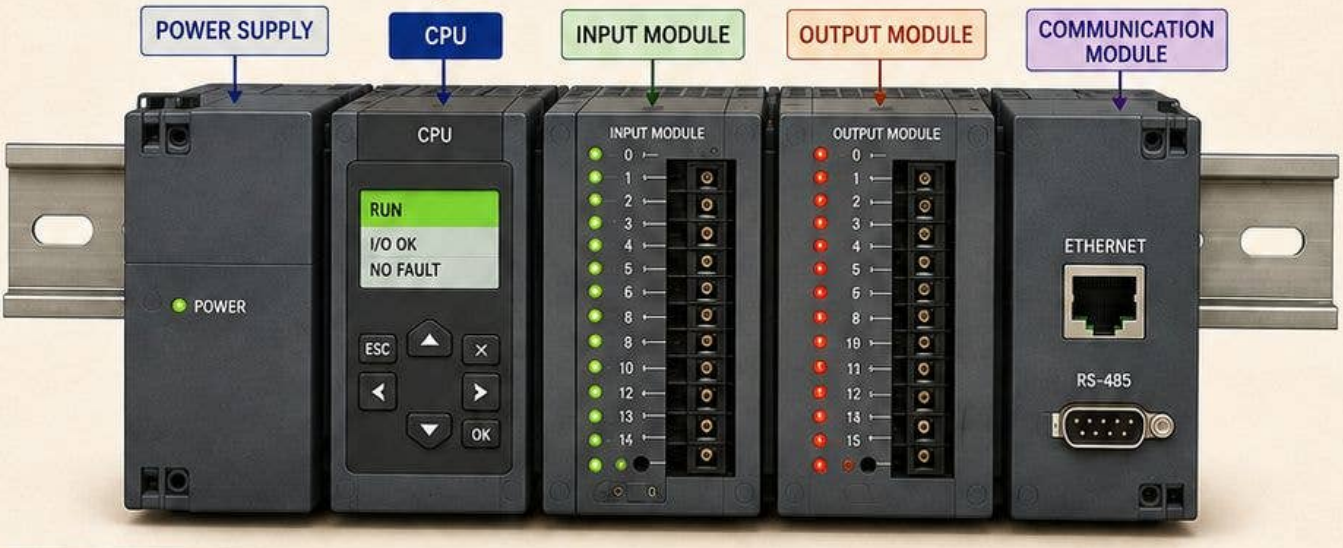
- Manufacturing automation
- Process control (chemical, food, pharma)
- Material handling systems
- Packaging machines
- Water treatment and pumping systems



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Topic 2

PLC Architecture



1. CPU (Central Processing Unit)

- The brain of the PLC.
- Executes user program.
- Performs logic operations.
- Controls all modules and data flow.
- Includes processor, registers, timers, counters.

2. Memory

- Stores program, data and system information.
- Types:
  - ROM / Flash (System)
  - RAM (Data & Program)
  - EEPROM (Retentive data)
- Holds I/O status, timers, counters, flags, etc.

3. Power Supply

- Converts AC to DC power for PLC and modules.
- Provides regulated voltage (e.g., 24V DC).
- Ensures stable and reliable operation of the system.

4. Input / Output Modules

- Interface between PLC and field devices.
- Input Modules: Read signals from switches, sensors, etc.
- Output Modules: Drive actuators, motors, valves, lamps, etc.
- Can be Digital or Analog.

5. Communication Interface

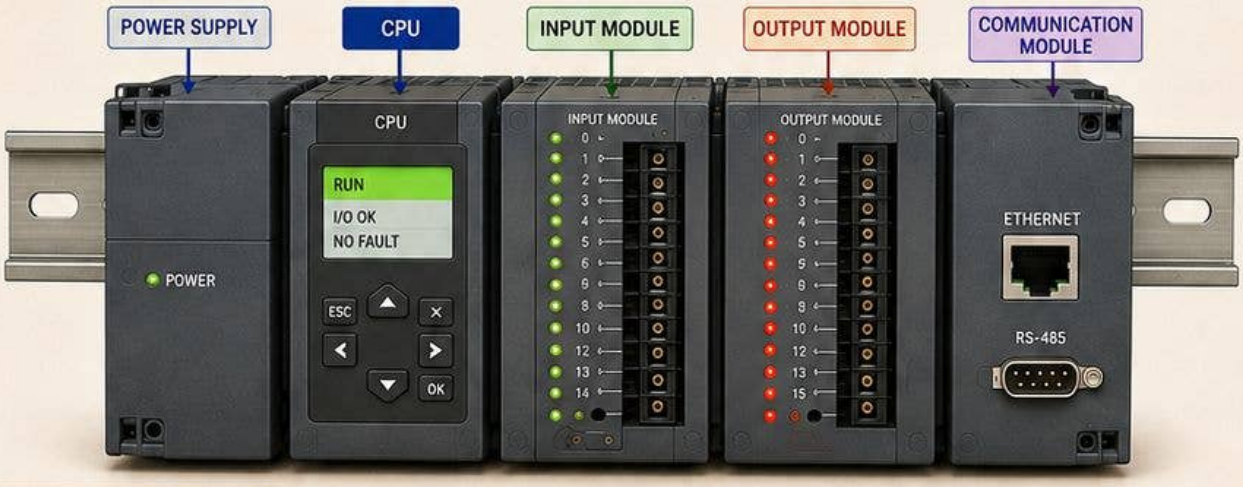
- Enables PLC to communicate with other devices and systems.
- Supports protocols like Modbus, Profibus, Ethernet/IP, CAN, etc.
- Used for HMI, SCADA, remote I/O, programming and networking.

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Topic 3

PLC

Inputs & Outputs



1. Digital I/O

- Handles ON/OFF signals.
- Input: Push buttons, limit switches, proximity switches, etc.
- Output: Relays, solenoids, lamps, buzzers, etc.
- Operates at fixed logic levels (0 or 1).



2. Analog I/O

- Handles continuous signals.
- Input: Temperature, pressure, level, flow, etc.
- Output: Speed control, position control, etc.
- Typically 0–10V, ±10V, 4–20mA signals.



3. Sensors (Input Devices)

- Detect physical conditions or events.
- Convert real-world signals into electrical signals for PLC.
- Examples: Proximity sensor, Photoelectric sensor, Temperature sensor, Pressure sensor, Level sensor.



4. Actuators (Output Devices)

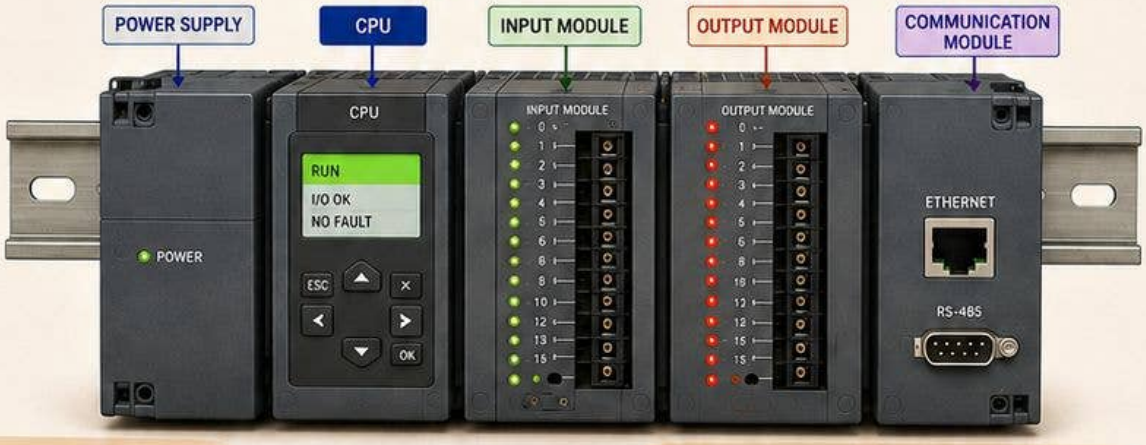
- Receive signals from PLC and perform actions.
- Convert electrical signals into physical output.
- Examples: Motor, Solenoid valve, Cylinder, Alarm, Heater.



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Topic 4

PLC
Scan Cycle



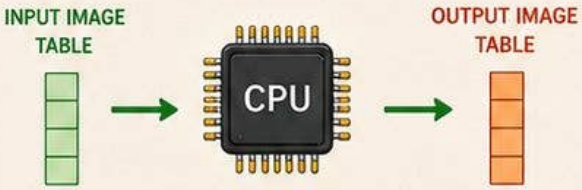
1. Input Scan

- PLC reads the status of all input devices.
- Input signals are stored in the input image table.
- This data is used by the program during execution.



2. Program Execution

- PLC executes the user program instructions.
- Uses the input data (from input image table) to solve logic.
- Results are written to the output image table.



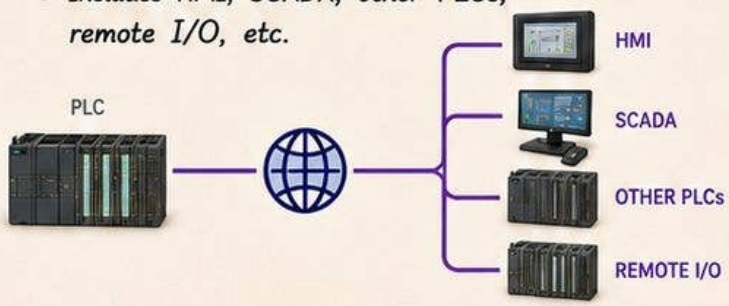
3. Output Update

- PLC transfers data from the output image table to output modules.
- Output modules energize the connected output devices.
- Changes in outputs take effect after this stage.



4. Communication

- PLC handles communication with other devices and systems.
- Exchanges data via communication modules and protocols.
- Includes HMI, SCADA, other PLCs, remote I/O, etc.



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Topic 5

Ladder Logic Fundamentals

1. NO & NC Contacts

- NO contact passes logic when input is ON.
- NC contact passes logic when input is OFF.



2. Coils

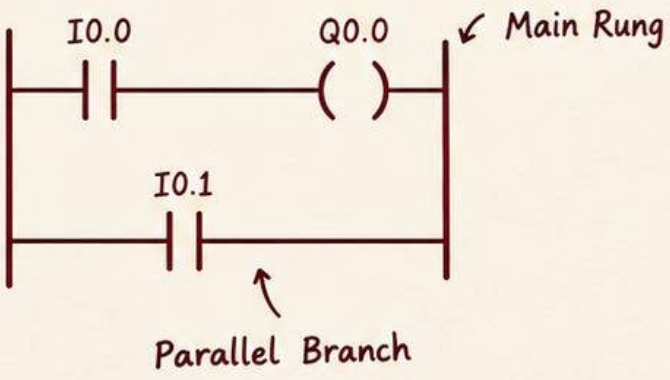
- Coil turns an output ON when the rung is TRUE.



Output Coil

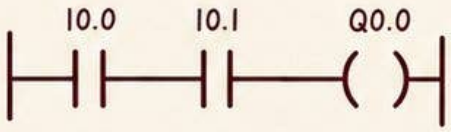
3. Rungs & Branches

- A rung is one horizontal logic line.
- Branches create parallel logic paths.

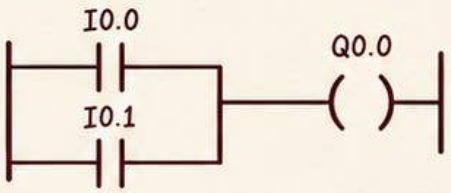


4. Basic Logic

- **AND** (series)



- **OR** (parallel)



- **NOT** (inversion)



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PLC

Topic 6

Timers & Counters

1. TON (On Delay Timer)

- Starts timing when input is ON.
- Output turns ON after preset time.
- Resets when input is OFF.



(Starts after 5 seconds)

2. TOF (Off Delay Timer)

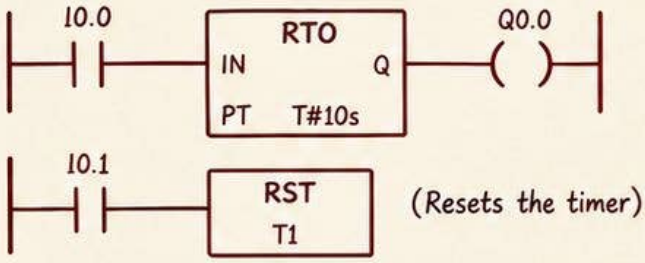
- Starts timing when input goes OFF.
- Output stays ON for preset time.
- Resets when input is ON again.



(Stays ON for 5 seconds after OFF)

3. Retentive Timer (RTO)

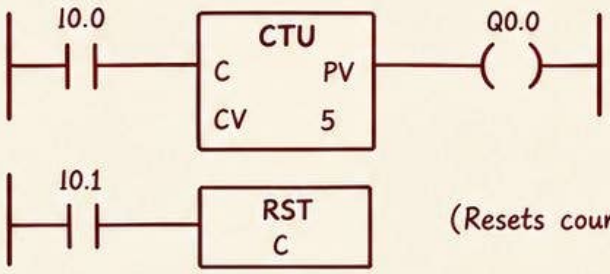
- Retains elapsed time even if input goes OFF.
- Only resets when RST is activated.



(Resets the timer)

4. CTU (Count Up)

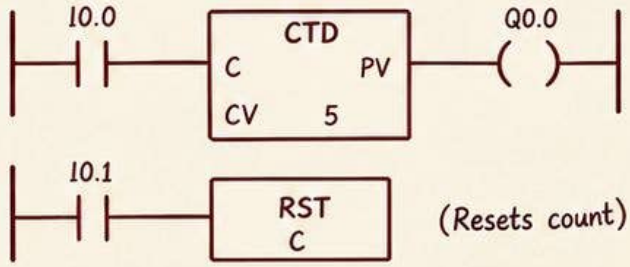
- Counts when input goes from OFF to ON.
- Reaches preset value (PV) and sets Q.
- Resets when R input is ON.



(Resets count)

5. CTD (Count Down)

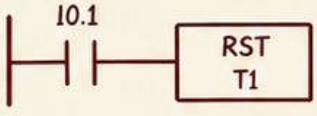
- Counts down when input goes from OFF to ON.
- Reaches 0 and sets Q.
- Resets when R input is ON.



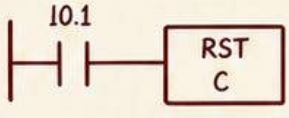
(Resets count)

6. Reset (R)

- Used to reset timers or counters.
- When R input is ON, the value is cleared.



(Reset timer)



(Reset counter)



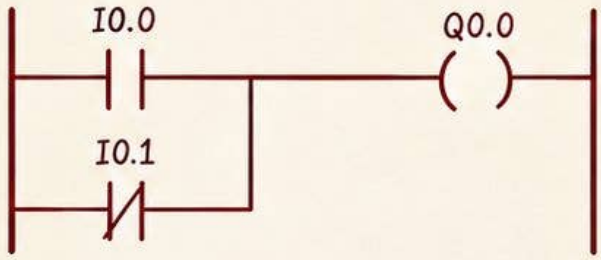
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PLC Programming Languages

Topic 7

1. Ladder Diagram (LD)

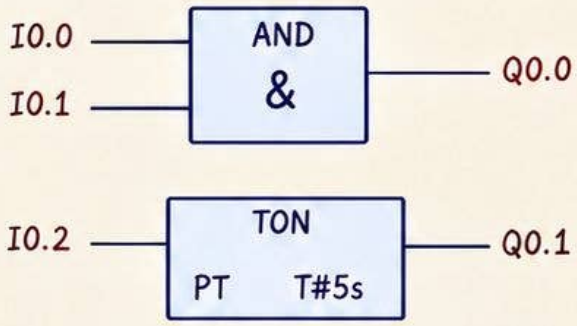
- Uses symbols like contacts and coils to represent logic.
- Looks like an electrical ladder.
- Widely used and easy to read.



(NO and NC contacts with coil)

2. Function Block Diagram (FBD)

- Uses function blocks and connections.
- Represents logic with graphical blocks.
- Good for complex logic and math functions.



(Function blocks connected with signals)

3. Structured Text (ST)

- Uses high-level programming language.
- Similar to languages like C.
- Good for complex calculations and logic.

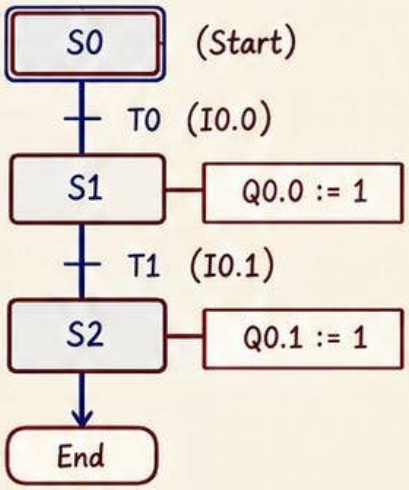
```
IF I0.0 AND I0.1 THEN
    Q0.0 := TRUE;
ELSE
    Q0.0 := FALSE;
END_IF;

TIMER(IN := I0.2, PT := T#5s);
Q0.1 := TIMER.Q;
```

(Text based program)

4. Sequential Function Chart (SFC)

- Uses steps and transitions.
- Shows the sequence of operations.
- Useful for process and machine control.



(Steps and transitions)

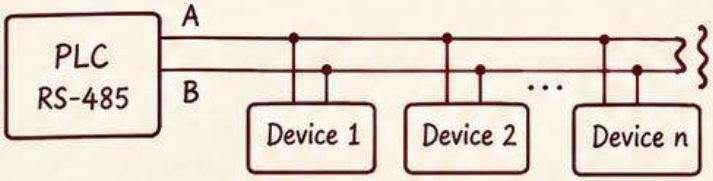
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PLC Communication

Topic 8

1. RS-485

- Uses differential signalling (A & B).
- Supports multi-drop (many devices).
- Long distance & high noise immunity.
- Common in industrial environments.



(Multi-drop communication)

2. Modbus

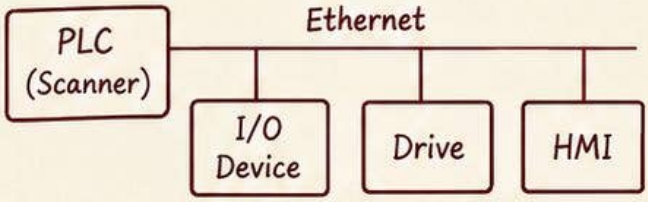
- Open, simple, and widely used protocol.
- Works over RS-232, RS-485, Ethernet (TCP/IP).
- Master-Slave communication.
- Common function codes (read/write).



Example: Read Holding Registers (Function 03)

3. Ethernet/IP

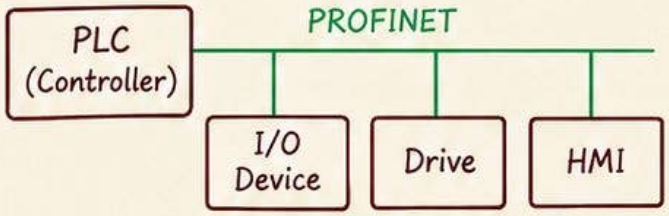
- Uses standard Ethernet (TCP/IP).
- Supports real-time and non-real-time data.
- Common in Rockwell Automation systems.
- Uses CIP (Common Industrial Protocol).



(CIP over TCP/IP)

4. PROFINET

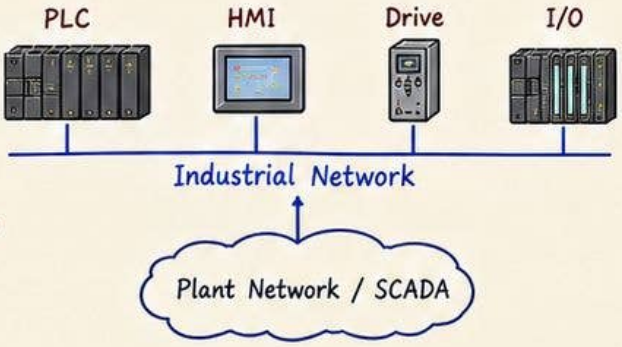
- Based on Industrial Ethernet.
- Real-time, high speed, deterministic.
- Used in Siemens systems.
- Supports IO, drives, and HMI.



(Real-time communication)

5. Industrial Networking

- Connects PLCs, HMIs, drives, I/O, and other devices.
- Enables data exchange and centralized control.
- Uses protocols like Modbus, PROFINET, EtherNet/IP, etc.
- Improves automation, monitoring and efficiency.



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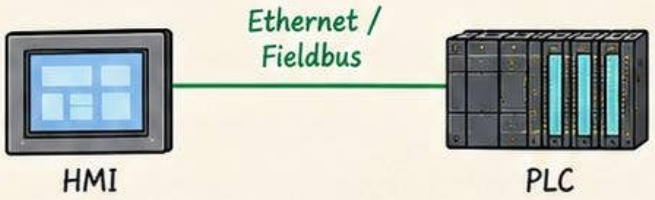
Topic 9

HMI, SCADA & PLC

Monitoring • Control • Alarms • Data Logging

1. PLC – HMI Connection

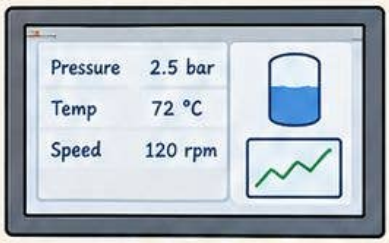
- HMI communicates with PLC using industrial protocols (e.g., Modbus TCP/IP, Ethernet/IP, PROFINET).
- Connected through Ethernet / switch or serial (RS-485).
- HMI reads and writes data to/from PLC.



(HMI ↔ PLC connection)

2. Monitoring

- HMI/SCADA reads real-time data from PLC.
- Displays process variables, status, trends and equipment health.
- Helps the operator keep track of the system.

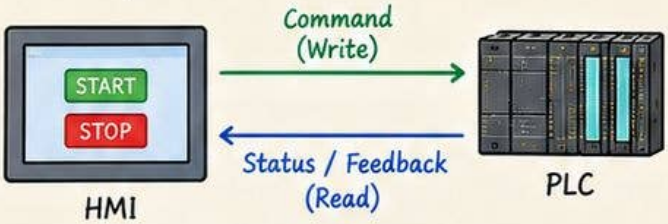


HMI / SCADA Screen

- Real-time values
- Trends
- Equipment status
- Process overview

3. Control

- Operator can send commands from HMI/SCADA to PLC.
- Changes setpoints, start/stop, mode, etc.
- PLC executes the logic and controls the process.



4. Alarms

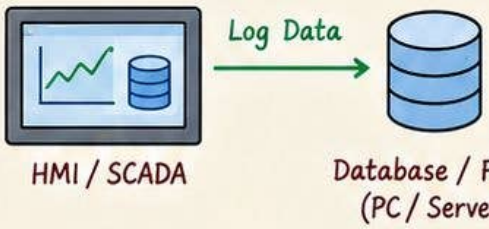
- HMI/SCADA shows alarm messages and alerts.
- Based on limit violations or fault conditions from PLC.
- Helps in quick response and safe operation.



- Visual (pop-ups, color change)
- Audio (buzzer)
- Event log

5. Data Logging

- HMI/SCADA stores process data over time.
- Used for analysis, reporting and traceability.
- Stored in files, database or cloud.
- Helps in performance improvement and maintenance.



- Trends
- Reports
- Analysis
- Traceability

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Topic 10

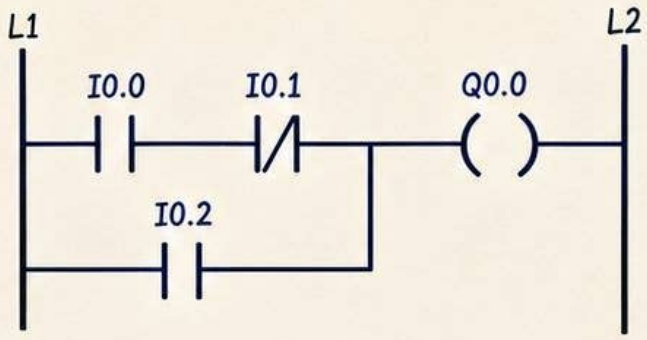
Ladder Logic Fundamentals

1. NO/NC Contacts & Coils

- NO (Normally Open) contact is closed when the condition is true.
- NC (Normally Closed) contact is closed when the condition is false.
- Coil is used to turn ON/OFF an output when the logic is true.



2. Ladder Diagram Example

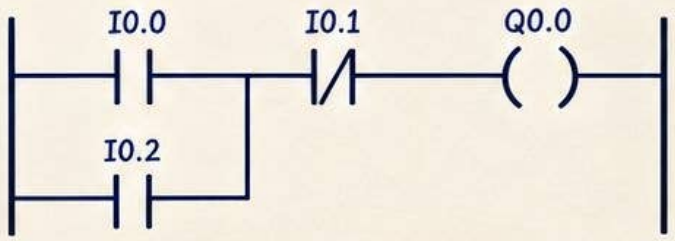


- Each rung represents a logic operation.
- Contacts and coils are placed from left to right.

3. Rungs & Branches

- **Rung:**
  A horizontal line where logic is written using contacts and coils.
- **Branch:**
  Used to create parallel paths (OR condition).

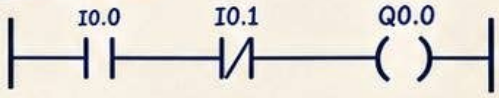
Rung with Branch



4. Basic Logic

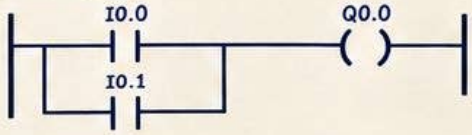
- AND Logic – all conditions must be true.
- OR Logic – any one condition can be true.
- NOT Logic – inverts the signal (using NC contact).

AND Logic



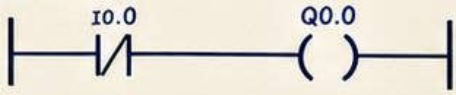
(both must be true)

OR Logic



(either one can be true)

NOT Logic



(true when input is false)